	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	598 - multibit_t	(782, 1, 1)x(256, 1, 1)	18,21 us	26 936	0 - NVIDIA GeForce RTX 3050 Ti Laptop GPU	1,48 Ghz	[20148] gpu_multibit_trie_stride4_REAL_TRAFFIC_V5.exe	

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

Command line profiler metrics

dram__throughput.avg.pct_of_peak_sustained_elapsed [%]	46,42	gpu__time_duration.sum [us]	18,21
lts__t_sector_hit_rate.pct [%]	55,27	sm__throughput.avg.pct_of_peak_sustained_elapsed [%]	46,00
smsp__warp_issue_stalled_long_scoreboard_per_warp_active.pct [%]	66,35	-	-

GPU Speed Of Light Throughput

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor.

Compute (SM) Throughput [%]	46,00	Duration [us]	18,21
Memory Throughput [%]	46,42	Elapsed Cycles [cycle]	26 936
L1/TEX Cache Throughput [%]	52,87	SM Active Cycles [cycle]	21 996,60
L2 Cache Throughput [%]	15,73	SM Frequency [Ghz]	1,48
DRAM Throughput [%]	46,42	DRAM Frequency [Ghz]	5,98

Latency Issue This workload exhibits low compute throughput and memory bandwidth utilization relative to the peak performance of this device. Achieved compute throughput and/or memory bandwidth below 60.0% of peak typically indicate latency issues. Look at Scheduler Statistics and Warp State Statistics for potential reasons.

Key Performance Indicators

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	782	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	16	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	200 192	Driver Shared Memory Per Block [Kbyte/block]	1,02
Waves Per SM	6,52	Shared Memory Configuration Size [Kbyte]	8,19
Uses Green Context	0	Stack Size	1 024
# SMs [SM]	20	# TPCs	10
Enabled TPC IDs	all	-	-

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	16
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	8
Achieved Occupancy [%]	72,92	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	35,00	Block Limit SM [block]	16

Achieved Occupancy
Est. Local Speedup: 27.08%

The difference between calculated theoretical (100.0%) and measured achieved occupancy (72.9%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Key Performance Indicators

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	21 996,60	Average L1 Active Cycles [cycle]	21 996,60
Average L2 Active Cycles [cycle]	20 394,88	Average SMSP Active Cycles [cycle]	21 286,51
Average DRAM Active Cycles [cycle]	50 500	Total SM Elapsed Cycles [cycle]	505 670
Total L1 Elapsed Cycles [cycle]	505 670	Total L2 Elapsed Cycles [cycle]	410 432
Total SMSP Elapsed Cycles [cycle]	2 022 680	Total DRAM Elapsed Cycles [cycle]	435 200

SMs Workload Imbalance
Est. Speedup: 6.89%

One or more SMs have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 7.92% above the average, while the minimum instance value is 4.17% below the average.

Key Performance Indicators

SMSPs Workload Imbalance
Est. Speedup: 8.02%

One or more SMSPs have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 9.53% above the average, while the minimum instance value is 5.78% below the average.

Key Performance Indicators

L1 Slices Workload Imbalance
Est. Speedup: 6.89%

One or more L1 Slices have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 7.92% above the average, while the minimum instance value is 4.17% below the average.

Key Performance Indicators

To customize your report even further, you might want to learn about custom sections and writing your own rules.
You might also want to consider adding individual metrics.

You can ask Nsight Copilot questions about this report.